

SIMATS ENGINEERING

ASSESSMENT TOOL 3

Comparative Analysis

Course Name: Computer Networks

Course Code: CSA07

Course Outcome: CO4 — Design network topologies (star, mesh, bus, and hybrid) using networking devices, transmission media, and cabling standards

Assessment Weightage: 30%

Total Marks: 25 (5 Questions)

Code of Conduct

I, C KARTHIK REDDY(192425157), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: C. Karthik Reddy

Comparative Analysis – Answers

Q1. Star vs Bus Topology — Cost, Reliability, and Performance

Cost Comparison

Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install.

Reliability Comparison

In terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network.

Performance Comparison

Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.

Scalability and Future Growth

Star topology scales far more gracefully: adding a new device is as simple as running a cable to a free switch port, with no impact on existing users. Bus topology, however, has a hard ceiling — every additional device increases contention for the shared medium, and beyond a certain point (typically a few dozen nodes on legacy Ethernet) collisions and retransmissions degrade throughput sharply, making it unsuitable for growing organisations.

Real-World Example

A modern office LAN is almost always built as a star: employees plug into wall ports that lead back to a wiring closet switch, and IT staff can isolate a faulty cable without affecting anyone else. Legacy bus-based Ethernet (10BASE2/10BASE5) is now largely obsolete outside of teaching labs and legacy industrial control systems, precisely because its single-point-of-failure backbone and poor scalability could not keep pace with growing bandwidth and reliability demands.

Which to Choose

Star topology is the clear choice whenever budget allows and reliability matters, which covers almost all modern business networks. Bus topology remains justifiable only where cost is the dominant constraint and the network is small, static, and non-critical, such as a temporary test bench or a very small legacy lab.

Cabling and Physical Media

Star topology typically uses twisted-pair copper cabling (Cat5e/Cat6) running from each device to a central switch, and because each run is independent, damaged or worn cable segments are cheap and easy to replace

without any wider network impact. Bus topology historically relied on coaxial cable (RG-58) with T-connectors and terminating resistors at each end; this shared medium is more fragile in practice, since a poor connection or damaged connector anywhere along the backbone can degrade or halt communication for every device on the segment.

Bandwidth Utilisation

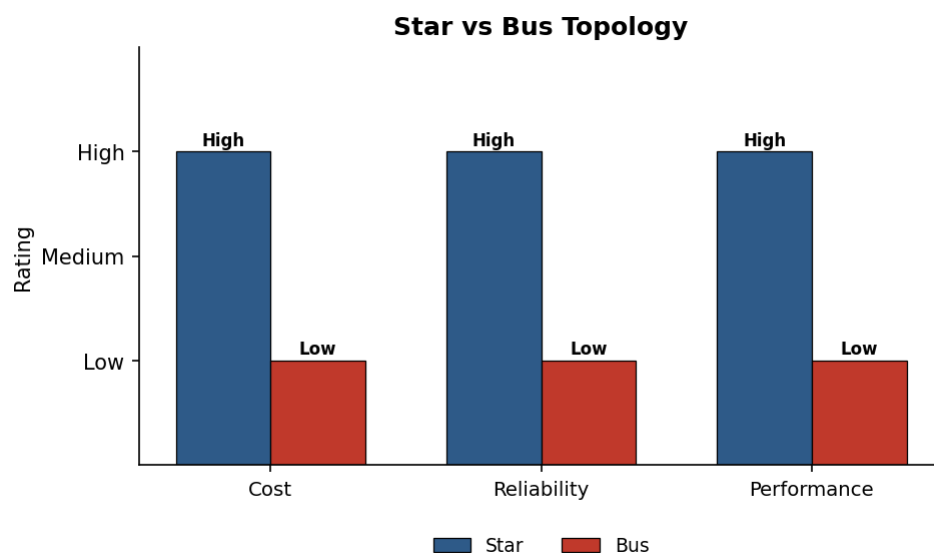
In a star network, the central switch can provide full, dedicated bandwidth to each connected port simultaneously (particularly with a switch rather than a hub), so multiple conversations can occur at once without interfering with each other. In a bus network, all devices share the same physical channel, so total available bandwidth must be divided among active devices, and only one device can transmit successfully at a time — any simultaneous transmission results in a collision that both senders must detect and retry.

Security Considerations

Star topology offers better security because traffic to each device can be isolated and monitored at the switch level, and features like port security, VLANs, and access control lists can be applied per connection. Bus topology is inherently less secure because every device shares the same physical medium, meaning that with the right equipment, traffic intended for one node could potentially be intercepted by another device attached to the same backbone.

Summary Verdict

Weighing cost against reliability, performance, cabling durability, bandwidth efficiency, and security together, star topology is the stronger overall design for almost any organisation beyond the smallest and most budget-constrained deployments, and this is reflected in its near-universal adoption in modern wired LANs, while bus topology today survives mainly as a historical reference point and an occasional teaching example.



Q2. Mesh vs Star Topology — Fault Tolerance

Mesh Topology Fault Tolerance

Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication.

Star Topology Fault Tolerance

In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning.

Cost of Fault Tolerance

The higher resilience of mesh topology comes at a significant price: every additional link requires extra network interfaces, cabling, and switch ports, so the number of connections grows quadratically with the number of nodes. Star topology avoids this overhead entirely, since redundancy would only need to be added at the single central device (for example, a second switch or a spare uplink), which is far cheaper than fully meshing every node.

Recovery and Routing Behaviour

In a mesh network, dynamic routing protocols such as OSPF or BGP continuously monitor link status and recalculate the best path within milliseconds of a failure, so end users typically notice no interruption. A star network has no such alternate-path mechanism; if the central switch fails, every attached device loses connectivity simultaneously until the switch is repaired or replaced, making recovery entirely dependent on hardware redundancy or a fast swap.

Conclusion

Therefore, mesh topology is more robust and reliable in critical environments compared to star topology, and is generally reserved for core layers of networks — such as bank data centres or ISP backbones — where the cost of downtime far outweighs the extra cabling expense, while star topology remains the practical choice at the access layer where a single switch failure affects a smaller, more easily managed group of users.

Latency and Path Selection

Mesh networks can also select the shortest or lowest-latency path among several available routes, which is valuable for time-sensitive traffic such as financial transactions or voice calls. Star networks have only one possible path between any two devices — through the central switch — so there is no opportunity for path optimisation; the only way to improve latency is to upgrade the switch itself or reduce the number of hops elsewhere in the network.

Impact of Scale on Fault Tolerance

As a mesh network grows, its fault tolerance advantage becomes even more pronounced, since more nodes mean more alternate paths are available when any single link fails. Conversely, as a star network grows by attaching more devices to the same central switch, the impact of that switch failing becomes worse, not better, because a

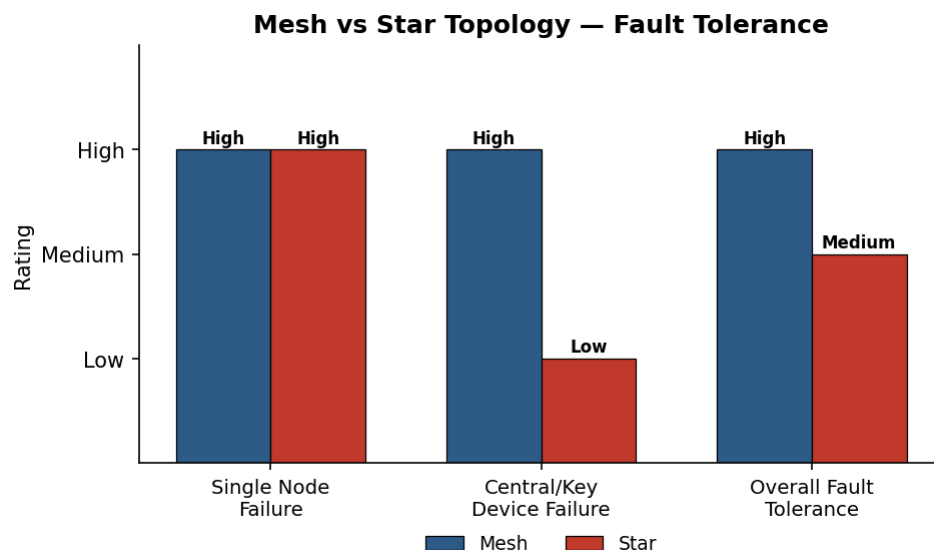
larger number of dependent devices would be affected simultaneously — meaning the fault tolerance of a star network effectively degrades as it scales, while a mesh network's fault tolerance can improve.

Practical Deployment Pattern

In practice, banks and other mission-critical organisations rarely deploy a pure mesh across their entire network, since the cost would be prohibitive; instead, they apply mesh connectivity only at the core layer among a small number of critical devices, while access and distribution layers use star topology with dual uplinks for redundancy. This hybrid approach captures most of the fault-tolerance benefit of mesh where it matters most, while keeping the overall cost and complexity manageable.

Overall Assessment

In summary, mesh topology decisively outperforms star topology on pure fault tolerance, but this advantage comes with substantially higher cost, complexity, and installation effort, meaning the right choice depends heavily on how critical continuous uptime is to the organisation in question.



Q3. Bus vs Mesh Topology — Installation Complexity

Bus Topology Installation

Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks.

Mesh Topology Installation

On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports.

Scalability of Complexity

This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology. Mathematically, a full mesh of n nodes requires $n(n-1)/2$ links — so a 5-node mesh needs 10 cables, but a 10-node mesh already needs 45, making full mesh impractical beyond a small number of core devices.

Planning and Skilled Labour

Installing a bus network requires little planning beyond laying the backbone cable and terminating both ends correctly, so it can typically be done by less experienced technicians in a short time. Mesh installation, by contrast, demands careful physical and logical planning — port allocation, cable labelling, and routing protocol configuration — and is usually carried out by experienced network engineers, further adding to project timelines and labour cost.

Partial Mesh as a Practical Compromise

Because full mesh is rarely practical outside small core layers, most real deployments use a partial mesh, connecting only the most critical devices (such as core routers and switches) to each other while less critical devices use star or hierarchical connections. This retains much of the fault tolerance of a full mesh while keeping installation complexity and cost much closer to that of simpler topologies like star or bus.

Time and Project Duration

Because of its simplicity, a small bus network can often be installed and made operational within a single working day, including running the backbone cable, attaching drop connections, and terminating both ends. A comparable mesh network of even modest size, by contrast, can take considerably longer to deploy, since every link must be individually run, tested, and documented, and the routing configuration on each device must be verified to ensure paths are being calculated and used correctly.

Testing and Verification

Testing a bus network is relatively straightforward: a technician can use a cable tester or time-domain reflectometer to check the single backbone for breaks or impedance mismatches, and confirm that the terminators are correctly seated at each end. Testing a mesh network is far more involved, since every individual link must be verified for connectivity and correct routing behaviour, and the network's failover behaviour (what happens

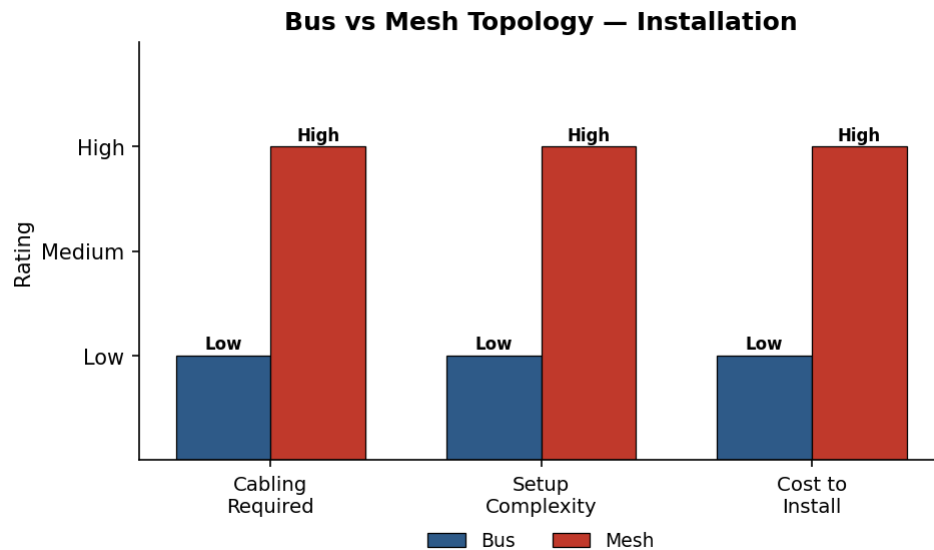
when a link is intentionally disconnected) should ideally be tested as well, which multiplies the verification effort considerably.

Documentation Requirements

A bus network's documentation needs are minimal — essentially a diagram of the backbone and a list of connected devices. A mesh network, however, requires detailed documentation of every link, port assignment, and routing protocol configuration, both to support ongoing operations and to make troubleshooting feasible when problems eventually arise; without this documentation, tracing a fault through dozens or hundreds of interconnections becomes extremely difficult.

Overall Assessment

Taking installation time, testing effort, required expertise, and documentation overhead together, bus topology is unambiguously simpler and faster to install than mesh topology, but this simplicity is achieved at the direct expense of the fault tolerance and performance benefits that make mesh topology attractive for critical infrastructure.



Q4. Hybrid vs Star Topology — Scalability

Hybrid Topology Scalability

Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure.

Star Topology Scalability

In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware.

Structural Reasons for the Difference

A pure star network has a single point of aggregation, so growth is fundamentally bounded by that one device's port count and backplane capacity; scaling beyond it means either replacing the switch with a larger one or stacking switches, which adds cost and complexity in an ad-hoc way. A hybrid network, by design, distributes this aggregation across multiple layers — for example, core, distribution, and access switches — so new capacity can be added at the layer where it is needed (a new access switch for a new floor, a new distribution switch for a new building) without redesigning the whole network.

Cost of Scaling Over Time

Because hybrid designs anticipate growth through a hierarchical structure, the incremental cost of adding capacity tends to be predictable and proportional to the growth itself. Star networks that outgrow their central device often face a disruptive and expensive forklift upgrade, where the core switch must be replaced entirely, causing downtime and requiring re-cabling or reconfiguration of existing connections.

Conclusion

Thus, hybrid topology provides greater scalability than star topology, and is generally preferred for organisations that expect sustained growth across multiple locations or departments, while a simple star topology remains adequate for smaller, single-site networks that are not expected to expand significantly beyond their initial design.

Geographic Expansion

Hybrid topology handles geographic expansion naturally, since a new building or campus can simply be connected to the core layer via a new distribution switch and fibre uplink, without requiring any changes to existing buildings' internal networks. Star topology has no inherent mechanism for geographic expansion at all; extending a pure star network across multiple buildings would require impractically long cable runs back to a single central switch, which is precisely why real-world large deployments almost always evolve into some form of hybrid design as they grow.

Redundancy Options During Growth

As a hybrid network scales, redundancy can be added incrementally at whichever layer needs it most — for example, adding a second core router for the whole campus, or a second distribution switch for just one busy

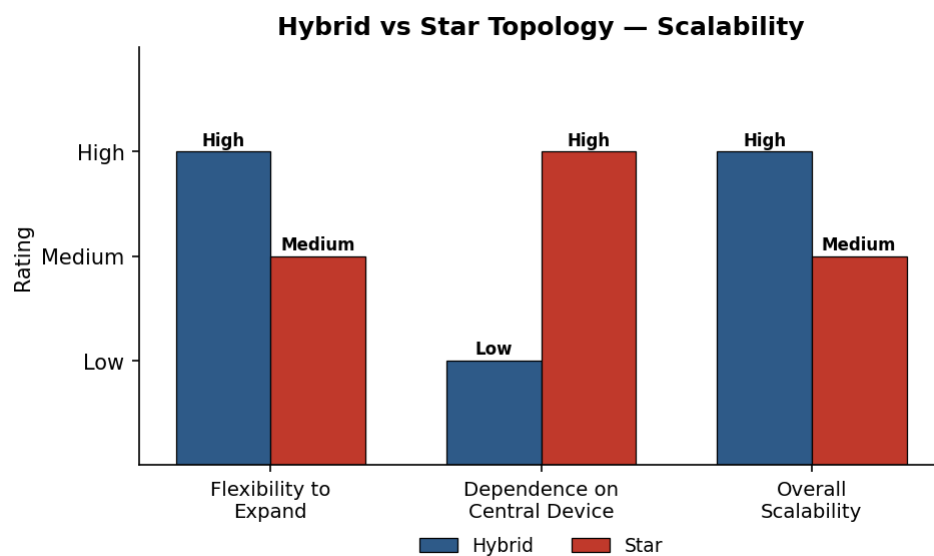
building — allowing investment to be targeted where it delivers the most benefit. In a star network, the only redundancy option is duplicating the single central device, which protects the whole network uniformly but offers no way to prioritise investment toward the areas of highest risk or highest usage.

Administrative Complexity of Scaling

It is worth noting that hybrid topology's superior scalability does come with greater administrative complexity: network administrators must manage multiple layers, routing between them, and a more elaborate IP addressing and VLAN scheme. Star topology remains administratively simpler throughout its usable range, which is part of why it continues to be the right choice for small and medium networks that do not need to scale much beyond a single wiring closet.

Overall Assessment

Balancing growth potential against administrative overhead, hybrid topology is the stronger choice for any organisation that anticipates expanding across multiple buildings, departments, or sites, while star topology remains a perfectly sound and simpler choice for organisations whose size and structure are expected to stay relatively stable.



Q5. Maintenance Comparison — Star, Mesh, Bus, and Hybrid

Star Topology Maintenance

In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable.

Mesh Topology Maintenance

Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming.

Bus Topology Maintenance

Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate.

Hybrid Topology Maintenance

Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

Tools and Practices That Ease Maintenance

Regardless of topology, good maintenance practice relies on network monitoring software (such as SNMP-based tools, Nagios, or PRTG) that can alert administrators to failures quickly, structured cabling with clear labelling, and up-to-date network diagrams. These practices reduce the practical maintenance burden of more complex topologies like mesh and hybrid, though they can never fully offset the extra effort required simply because there are more devices and links to track.

Overall Ranking

Taking all four topologies together, star is the easiest to maintain due to its simple, centralised structure; hybrid is next, since its layered design keeps most faults localised to a single layer or segment even though the overall system is more complex; bus and mesh are the most demanding, but for opposite reasons — bus because a single shared cable makes faults hard to isolate, and mesh because the sheer number of interconnections makes tracing the source of a problem time-consuming, even though the network itself keeps running while the fault is diagnosed.

Diagnostic Approach for Each Topology

Diagnosing a star network fault typically starts and ends at the affected device's cable and switch port, since the star structure itself confines the problem's scope. Diagnosing a bus network fault usually requires physically walking the length of the backbone cable with a tester to locate a break or bad connector, since there is no way to isolate individual segments electronically. Diagnosing a mesh fault requires reviewing routing tables and link status across many devices to determine which specific link has failed, even though the overall network typically continues functioning while this investigation takes place. Diagnosing a hybrid fault requires first identifying

which layer (core, distribution, or access) is affected, then applying the appropriate one of the above approaches to that layer specifically.

Staffing and Skill Requirements

Star and simple hybrid access-layer maintenance can generally be handled by junior IT staff or helpdesk technicians, since faults are localised and diagnostic steps are straightforward. Mesh and core-layer hybrid maintenance typically requires more senior network engineers with a solid understanding of routing protocols, since misdiagnosing or mishandling a core fault can have network-wide consequences. Bus topology maintenance sits in between: it does not require deep protocol knowledge, but does require patience and appropriate cable-testing equipment to physically locate faults.

Downtime During Maintenance

Scheduled maintenance on a star network (such as replacing a switch) affects only the devices connected to that switch, and can often be done outside business hours with minimal disruption. Maintenance on a bus network's backbone typically requires taking the entire segment offline, since there is no way to work on the shared cable while it remains in use. Mesh networks offer the best maintenance experience in this respect, since individual links or devices can usually be taken down for maintenance one at a time while traffic automatically reroutes through the remaining paths, allowing maintenance with little or no visible downtime.

Long-Term Maintenance Cost

Over the lifetime of a network, star topology tends to have the lowest ongoing maintenance cost due to its simplicity and the availability of low-cost replacement parts and less specialised labour. Mesh topology has the highest ongoing maintenance cost, driven by the need for skilled engineers, more complex diagnostic tools, and a larger quantity of cabling and interfaces that can individually fail over time. Bus and hybrid topologies fall in between, with bus being cheap per fault but slow to diagnose, and hybrid balancing moderate skill requirements against the benefit of faults typically being confined to a single layer.

Final Recommendation

For organisations with limited IT staff and a preference for predictable, low-cost maintenance, star topology (or a simple hybrid built mostly from star segments) is the most practical choice. Organisations that can justify the higher maintenance investment — typically those where downtime is extremely costly, such as banks, hospitals, or large data centres — are better served by incorporating mesh connectivity at the core, accepting the additional skilled labour and diagnostic overhead as the price of continuous availability.

Ease of Maintenance by Topology

